

# Structural Plane Stress from Generic Three-Dimensional MFront Behaviours

A constitutive closure for small-strain implicit StandardElasticity-compatible laws

Kinematics-Driven 316L Strain Reconstruction

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### Category: scientific reference and upstream design note.

This note describes a structural plane-stress closure for a genuinely three-dimensional constitutive behaviour. It is not a plate or shell theory: the finite-element model is a two-dimensional membrane model with a local constitutive closure at each material point.

The demonstrated V1 domain is deliberately narrow:

small strain + Implicit + Tridimensional + StandardElasticity-compatible elastic split.

The constitutive state remains three-dimensional. The generic mathematical transformation has been demonstrated on rotated elasticity, J2, SRIX and Méric–Cailletaud; the currently registered full-field implementation still uses SRIX-specific source-generation scaffolding, documented in Section 12.

## 1. Structural assumptions and the meaning of plane stress

For a body with structural thickness coordinate  $z$ , a free face has the boundary condition

$$\sigma \mathbf{n} = \mathbf{0} \quad \text{on } z = +h/2 \text{ and } z = -h/2.$$

The two-dimensional plane-stress approximation extends this surface condition to the representative constitutive state through the thickness of a sufficiently thin membrane. Under that approximation, with  $\mathbf{n} = \mathbf{e}_z$ , the local closure is

$$\boxed{\sigma_{zz} = 0, \quad \sigma_{xz} = 0, \quad \sigma_{yz} = 0.}$$

This is an approximation to a three-dimensional boundary-value problem, not a claim that the traction condition holds at every point of a finite-thickness body. Bending, transverse shear and through-thickness variation are outside this formulation.

The FEM unknowns remain only

$$u_x(x, y), \quad u_y(x, y).$$

The three transverse strains

$$\varepsilon_b = (\varepsilon_{zz}, \gamma_{xz}, \gamma_{yz})$$

are local constitutive unknowns. They are solved and condensed at a material point; they are not additional FEM degrees of freedom.

## 2. Relation to existing MFront plane-stress hypotheses

MFront and `StandardElasticity` already provide powerful support for the standard `PlaneStress` hypothesis and also provide axial-stress relaxation machinery for the axisymmetrical generalised-plane-stress case. For `PlaneStress`, an axial-strain unknown is introduced and  $\sigma_{zz} = 0$  is enforced. In the axisymmetrical generalised-plane-stress case, the same axial strain is solved while the axial stress may be prescribed through an external state variable. These standard mechanisms remain axial: they do not introduce the two transverse shear strains required by the structural relaxation considered here. This note does not use `GeneralisedPlaneStress` as the name of a generic Cartesian MFront modelling hypothesis.

The problem addressed here is different:

| Existing standard mechanism                           | Structural 3D closure in this note             |
|-------------------------------------------------------|------------------------------------------------|
| additional local unknown typically $\varepsilon_{zz}$ | $\varepsilon_{zz}, \gamma_{xz}, \gamma_{yz}$   |
| closure typically $\sigma_{zz} = 0$                   | $\sigma_{zz} = \sigma_{xz} = \sigma_{yz} = 0$  |
| behaviour represented under a plane-stress hypothesis | behaviour remains <code>Tridimensional</code>  |
| suitable for the standard decoupled case              | suitable for arbitrary 3D material orientation |

The proposal does not replace MFront's existing plane-stress mechanisms. It targets a three-dimensional law with a complete three-dimensional internal state and a structural surface orientation that is independent of the crystal orientation. In particular, an EBSD crystal orientation is not generally aligned with the structural normal.

### 3. Tensor and Kelvin notation

The scientific derivation uses Kelvin components. The structural engineering interface uses engineering shear components. With component order

$$(xx, yy, zz, xy, xz, yz),$$

the engineering strain and Kelvin strain vectors are

$$[\varepsilon]_{eng} = [\varepsilon_{xx} \quad \varepsilon_{yy} \quad \varepsilon_{zz} \quad \gamma_{xy} \quad \gamma_{xz} \quad \gamma_{yz}]^T,$$

$$[\varepsilon]_K = [\varepsilon_{xx} \quad \varepsilon_{yy} \quad \varepsilon_{zz} \quad \sqrt{2}\varepsilon_{xy} \quad \sqrt{2}\varepsilon_{xz} \quad \sqrt{2}\varepsilon_{yz}]^T, \quad \gamma_{ij} = 2\varepsilon_{ij}.$$

The stress Kelvin vector is

$$[\sigma]_K = [\sigma_{xx} \quad \sigma_{yy} \quad \sigma_{zz} \quad \sqrt{2}\sigma_{xy} \quad \sqrt{2}\sigma_{xz} \quad \sqrt{2}\sigma_{yz}]^T.$$

Let  $Q \in SO(3)$  map global/structural tensors to material tensors:

$$A_m = QA_gQ^T, \quad A_g = Q^T A_m Q.$$

The corresponding Kelvin operator is a distinct six-dimensional object,  $\mathcal{R}_K(Q) \in \mathbb{R}^{6 \times 6}$ :

$$[A_m]_K = \mathcal{R}_K(Q)[A_g]_K, \quad [A_g]_K = \mathcal{R}_K(Q)^T[A_m]_K.$$

Thus no 3-by-3 matrix is ever multiplied directly by a six-component vector. For a stress vector,  $[\sigma_g]_K = \mathcal{R}_K(Q)^T[\sigma_m]_K$ . For a fourth-order tangent, both input and output frames transform:

$$C_g = \mathcal{R}_K(Q)^T C_m \mathcal{R}_K(Q).$$

The three implementations use these operators differently:

| route               | tangent frame operation                                                                                                        |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------|
| raw 3D condensation | rotate both tangent input and output:<br>$C_g = \mathcal{R}_K^T C_m \mathcal{R}_K$                                             |
| specialised GPS     | differentiate with respect to the structural gradient; the host rotates material stress rows and applies the in-plane contract |

|             |                                                                                               |
|-------------|-----------------------------------------------------------------------------------------------|
| route       | tangent frame operation                                                                       |
| generic SPS | construct the structural plane-stress tangent directly from the transformed residual/Jacobian |

GPS behaviours receive the structural gradient and own the crystal rotation through per-point  $Q_{ij}$  properties. Applying a raw-bridge rotation as well would rotate the gradient twice.

#### 4. Why $\sigma_{zz} = 0$ alone fails for rotated anisotropy

Partition the structural Kelvin components as

$$a = (xx, yy, xy), \quad b = (zz, xz, yz).$$

The linearized response is

$$\begin{bmatrix} \sigma_a \\ \sigma_b \end{bmatrix} = \begin{bmatrix} C_{aa} & C_{ab} \\ C_{ba} & C_{bb} \end{bmatrix} \begin{bmatrix} \varepsilon_a \\ \varepsilon_b \end{bmatrix}.$$

For a rotated anisotropic material,  $C_{ba} \neq 0$  in general. A purely in-plane strain therefore produces transverse shear traction. Fixing  $\gamma_{xz} = \gamma_{yz} = 0$  and adjusting only  $\varepsilon_{zz}$  can make  $\sigma_{zz} = 0$  while leaving  $\sigma_{xz}$  and  $\sigma_{yz}$  non-zero.

Crystal plasticity makes the same point independently of elasticity:

$$\Delta \varepsilon^p = \sum_s \Delta \gamma_s \mathbf{M}_s.$$

After an arbitrary EBSD orientation, each Schmid tensor can have all six structural components. The three transverse strains must therefore be solved so that the complete traction vector vanishes.

#### Numerical illustration

The rotated elastic probe uses the Bunge orientation  $[35^\circ, 20^\circ, 15^\circ]$ , and the cubic elastic constants used in the 316L probe,  $C_{11} = 197000$ ,  $C_{12} = 125000$  and  $C_{44} = 122000$  MPa, and the structural engineering strain

$$(\varepsilon_{xx}, \varepsilon_{yy}, \gamma_{xy}) = (10^{-3}, -2 \times 10^{-4}, 3 \times 10^{-4}).$$

The same constitutive calculation gives:

| local closure         | $\varepsilon_{zz}$          | $\gamma_{xz}$ | $\gamma_{yz}$ | $\sigma_{zz}$ MPa      | $\sigma_{xz}$ MPa | $\sigma_{yz}$ MPa |
|-----------------------|-----------------------------|---------------|---------------|------------------------|-------------------|-------------------|
| axial relaxation only | $-4.0418897 \times 10^{-4}$ | 0             | 0             | $-1.6 \times 10^{-14}$ | 22.7252           | -42.5920          |

| local<br>closure                                        | $\varepsilon_{zz}$       | $\gamma_{xz}$            | $\gamma_{yz}$           | $\sigma_{zz}$ MPa     | $\sigma_{xz}$ MPa     | $\sigma_{yz}$ MPa     |
|---------------------------------------------------------|--------------------------|--------------------------|-------------------------|-----------------------|-----------------------|-----------------------|
| struc-<br>tural<br>traction-<br>free<br>relaxa-<br>tion | $-4.8198 \times 10^{-4}$ | $-1.5775 \times 10^{-4}$ | $2.5121 \times 10^{-4}$ | $1.7 \times 10^{-15}$ | $2.6 \times 10^{-14}$ | $1.4 \times 10^{-14}$ |

The first row satisfies  $\sigma_{zz} = 0$  but leaves substantial transverse shear traction. The second row is the structural plane-stress closure. The values are generated by the rotated elastic probe, not inferred from the qualification residual alone.

The non-zero transverse strains are a local constitutive relaxation of transverse modes. They do not imply that the 2D FEM field contains an explicit through-thickness displacement or warping field. In the elastic case this is the stationarity condition of the reduced energy

$$W^{PS}(\varepsilon_a) = \min_{\varepsilon_b} W^{3D}(\varepsilon_a, \varepsilon_b), \quad \frac{\partial W}{\partial \varepsilon_b} = \sigma_b = 0.$$

For a positive-definite elastic potential, this minimum is well-defined. For an incremental dissipative law, the same idea is realised by the local implicit constitutive equations rather than by a global energy minimisation. No explicit through-thickness heterogeneity or warping is resolved; a material that varies through the thickness requires a 3D, layered or shell/multilayer model.

## 5. External 3D condensation: the independent reference

The backend `mfront-3d-condensed-plane-stress` leaves the behaviour unchanged. It solves the three local equations

$$\sigma_b(\varepsilon_a, \varepsilon_b) = 0$$

with an outer Newton iteration. Every trial is restored from one complete committed snapshot. After convergence, the constrained tangent is

$$C^{PS} = C_{aa} - C_{ab} C_{bb}^{-1} C_{ba}.$$

The inverse is not formed: a 3-by-3 linear system is solved. The bridge can use either a committed or tangent transverse predictor, monitors the conditioning of  $C_{bb}$ , and records the local closure iterations. This is the independent reference because it can use an unchanged compatible 3D law.

## 6. Specialised SRIX monolithic closure: the constructive proof

Before the generic transformation was attempted, the same idea was written explicitly in `Fcc316LForestRubinSrixGps`. The SRIX local system contains six elastic unknowns `de1[6]` and twelve slip unknowns `dg[12]`. Its three in-plane elastic rows impose the structural in-plane increment, while rows `zz`, `xz` and `yz` impose the three structural traction equations. The remaining SRIX flow and hardening equations are unchanged.

This specialised behaviour demonstrated that a real anisotropic crystal law can converge with the three transverse kinematics relaxed inside its local implicit Newton. It also provided a second implementation against which the generic `StructuralPlaneStress3D` behaviour could be compared on M20 and M100. It is not generic: its source knows the SRIX slip variables and must be maintained as a law-specific GPS variant. Its qualified host integration still uses selective substepping and, where necessary, the composite tangent.

The specialised route is therefore the constructive proof that the closure is practical for crystal plasticity; the generic route is the attempt to express the same row/Jacobian transformation without knowing SRIX variables.

## 7. Exact equivalence of external and monolithic closure

The equivalence is easiest to see before differentiating. Let  $x$  contain the local constitutive unknowns and write the raw 3D system as

$$K_a(x) - \varepsilon_a = 0, \quad K_b(x) - \varepsilon_b = 0, \quad G(x) = 0,$$

where  $G$  denotes all other constitutive equations. Plane stress adds

$$S_b(x) = 0.$$

The second row equation gives  $\varepsilon_b = K_b(x)$ . Eliminating it leaves

$$K_a(x) - \varepsilon_a = 0, \quad G(x) = 0, \quad S_b(x) = 0,$$

which is exactly the monolithic structural closure. Therefore, on the same committed state, increment, orientation and differentiable constitutive branch,

|                                                                  |
|------------------------------------------------------------------|
| external 3D condensation $\equiv$ monolithic structural closure. |
|------------------------------------------------------------------|

The Schur tangent and the monolithic implicit tangent are two eliminations of the same local equations, not two different physical models.

## 8. Generic MFront residual transformation

The specialised SRIX behaviour writes the closure equations explicitly. The generic prototype instead uses the standard elastic residual as an algebraic access point. In the current generated hook, `StandardElasticity` has already assembled a block of the form

$$f_e^{std} = K_m - g,$$

where  $g$  is the six-component gradient stored in the standard `deto` slot. Here  $g$  is the structural gradient supplied by the host; it must not be renamed  $\Delta\epsilon_m$ . Before the standard rows are consumed by Newton, the hook recovers

$$K_m = f_e^{std} + g.$$

This is an implementation-level algebraic extraction, not a statement that a structural gradient is a material-frame tensor. A future upstream interface should expose  $K_m$  or the elastic block directly instead of relying on the generated residual layout.

The recovered tensor is transformed with  $\mathcal{R}_K(Q)^T$ . The six elastic rows are replaced by

$$F_a = (K_g)_a - g_a, \quad F_b = \frac{(\sigma_g)_b}{S_{ref}},$$

while all other constitutive rows are left unchanged. All columns of the elastic-row Jacobian are transformed, including columns belonging to unknowns whose names and physical meanings are not known to the closure.

The current V1 Jacobian transformation additionally assumes

$$\sigma_m = D_m e_m^{el},$$

with no direct dependence of stress on another implicit variable. If, for example,  $\sigma = \sigma(e^{el}, d)$  with an implicit damage variable  $d$ , then  $\partial\sigma/\partial d \neq 0$  and the currently zeroed transverse columns would not be correct. This is part of the contract, not an accidental detail.

## 9. Residual scaling

The traction rows are scaled as

$$F_b = \sigma_b / S_{ref}, \quad S_{ref} > 0.$$

The root is unchanged, but the scaling affects local Newton conditioning. The current generated prototype uses the fixed value  $S_{ref} = 210000$  MPa. This is a prototype limitation, not a generic physical constant. An upstream implementation should derive the scale from a representative elastic modulus or expose it as a well-documented behaviour parameter.

## 10. One-step consistent tangent

At convergence, let

$$F(x, g_a) = 0, \quad A = \frac{\partial F}{\partial x}.$$

Since  $F_a = (K_g)_a - g_a$ ,

$$\frac{\partial F}{\partial g_a} = -E_a,$$

where the only non-zero entries of  $E_a$  are

$$(E_a)_{0,0} = 1, \quad (E_a)_{1,1} = 1, \quad (E_a)_{3,2} = 1.$$

Thus the local sensitivity is obtained from three right-hand sides:

$$AX = E_a, \quad X = \frac{\partial x}{\partial g_a}.$$

The implementation solves this system and never forms  $A^{-1}$ . If  $X_e$  denotes the elastic-state rows and  $D_m$  is the material-frame stress/elastic-strain tangent, first form the six-output operator

$$\hat{C}^{PS} = \mathcal{R}_K(Q)^T D_m X_e \in \mathbb{R}^{6 \times 3}.$$

Let  $S_a \in \mathbb{R}^{3 \times 6}$  select the structural stress components xx, yy and xy. The tangent returned to the two-dimensional solver is

$$C^{PS} = S_a \hat{C}^{PS} = S_a \mathcal{R}_K(Q)^T D_m X_e \in \mathbb{R}^{3 \times 3}.$$

The six-output form is useful for checking the transverse traction rows, but the Schur complement is the selected 3-by-3 operator. Thus

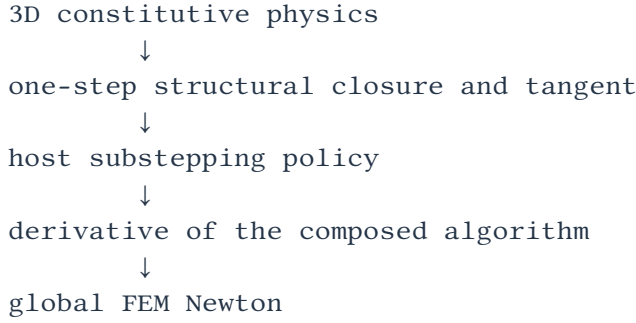
$$C^{PS} = C_{aa} - C_{ab} C_{bb}^{-1} C_{ba}.$$

The inactive structural columns are not independent input columns and are set to zero in the six-column operator exposed to the host. This is the tangent of one constitutive step.



## 11. Host substepping and the composite tangent

The implementation has three distinct derivative layers:



For a fixed sequence of maps, a simplified notation is

$$\Phi = \Phi_N \circ \dots \circ \Phi_1, \quad D\Phi = D\Phi_N \cdots D\Phi_1.$$

The actual driver distributes the target increment over the sub-steps,

$$\Delta g_k = \frac{1}{N}(g_{target} - g_n),$$

so each map also depends directly on the final target. With

$$F_k(z_k, S_{k-1}, \Delta g_k) = 0,$$

the sensitivity obeys the recurrence

$$A_k \frac{dz_k}{dg_{target}} = - \left( F_{S,k} \frac{dS_{k-1}}{dg_{target}} + F_{\Delta g,k} \frac{d\Delta g_k}{dg_{target}} \right).$$

The product formula is therefore only an illustration; the direct target terms and state propagation are part of the true composite derivative.

The qualified adapter reconstructs this derivative by central finite differences for points that actually sub-step. If the plus and minus perturbations follow the same partition, the result approximates the local derivative of the algorithmic map. If the adaptive partition changes, the map is only piecewise smooth and the finite difference is a secant/generalised Jacobian diagnostic rather than a classical derivative. Partition changes are recorded; constitutive branch changes are not currently detected generically.

| Control | External condensation | GPS / generic SPS |
|---------|-----------------------|-------------------|
|---------|-----------------------|-------------------|

## 12. Numerical controls and algorithmic choices

| Control                                    | External condensation                                   | GPS / generic SPS                                                 |
|--------------------------------------------|---------------------------------------------------------|-------------------------------------------------------------------|
| transverse predictor                       | committed or first-order $C_{bb}^{-1}C_{ba}$ predictor  | committed or previous accepted transverse-increment extrapolation |
| $C_{bb}$ conditioning                      | checked; policy always, on_failure or diagnostic_sample | not available from the constrained tangent                        |
| local closure iterations                   | outer Python transverse Newton                          | MFront implicit Newton                                            |
| selective substepping                      | not used by the reference closure                       | bisection, cached failure spans, maximum 256 subdivisions         |
| composite FD                               | not used                                                | gps_composite_fd_tangent, step 1e-6 absolute engineering strain   |
| shadow tangent                             | not applicable                                          | diagnostic only; selectable scope all, substepped, non_substepped |
| transverse tolerances and iteration limits | external closure controls                               | local MFront/host GPS controls                                    |
| MGIS thread count                          | parallel constitutive batch                             | parallel constitutive batch                                       |

The shadow tangent must not be enabled in production. The composite FD is sparse: on the qualified M100 run it touched 192 points and 1152 trajectories.

## 13. What is generic, and what is still prototype-specific

Three levels must be distinguished:

1. **Generic mathematical transformation.** The residual/Jacobian idea does not refer to SRIX, Méric, slips or hardening and has been demonstrated on rotated elasticity, J2, SRIX and Méric.
2. **Generic prototype closure/tangent.** The probes transform all local Jacobian columns and derive the local system size from the Jacobian type.
3. **Registered full-field backend.** The current generated SRIX backend still uses source-rewriting scaffolding, an exposed StructuralJacobian[324] auxiliary buffer and SRIX-oriented output extraction. Thus the industrialised field backend is not yet arbitrary-law generic. In particular,  $324=18^2$  is a current implementation limitation and a law with a different local system size is not yet covered by this production transport path.

The external Behaviour Brick route was investigated, but the installed TFEL 5.1 ABI does not export the required BehaviourBrickBase symbols. No TFEL fork is used. The honest current genericity claim is therefore:

The transformation is mathematically generic for the demonstrated V1 contract; the registered full-field implementation is currently qualified for SRIX.

## 14. Verification and qualification

The point-material evidence is:

| Behaviour                      | transverse traction        | in-plane kinematic error | tangent check                                                                 | what it proves                                  |
|--------------------------------|----------------------------|--------------------------|-------------------------------------------------------------------------------|-------------------------------------------------|
| rotated anisotropic elasticity | $2.3 \times 10^{-14}$ MPa  | $4.9 \times 10^{-19}$    | Schur $6.95 \times 10^{-16}$                                                  | frame transformation and elastic closure        |
| J2 plastic                     | $7.99 \times 10^{-14}$ MPa | —                        | FD $3.12 \times 10^{-11}$ at $h = 10^{-7}$                                    | all local columns can be transformed            |
| SRIX                           | $9.30 \times 10^{-15}$ MPa | $1.74 \times 10^{-18}$   | FD $2.48 \times 10^{-10}$ at $h = 10^{-7}$                                    | crystal plasticity closure and one-step tangent |
| Méric-Cailletaud               | $6.77 \times 10^{-15}$ MPa | $1.42 \times 10^{-19}$   | FD $3.97 \times 10^{-15}$ to $2.45 \times 10^{-13}$ over the tested $h$ range | second implicit crystal law, same closure code  |

The generic SRIX and Méric one-step tangents have been checked by finite differences. A dedicated, archived same-state comparison of those two generic variants against a live raw-3D Schur oracle is not yet available; the Schur identity is therefore a required next validation artifact, not a claim made by this table.

The P43 M100 single-run comparison used the same EBSD crop, eight increments, four MFront threads, one BLAS thread and one FFTW thread:

| strategy                                       | Newton | sequence                 | elapsed |
|------------------------------------------------|--------|--------------------------|---------|
| 3D + Python condensation                       | 57     | [6, 6, 7, 7, 7, 8, 8, 8] | 56.72 s |
| specialised GPS + composite FD                 | 58     | [6, 6, 7, 7, 7, 8, 8, 9] | 51.65 s |
| generic StructuralPlaneStress3D + composite FD | 58     | [6, 6, 7, 7, 7, 8, 8, 9] | 54.56 s |

The generic and specialised GPS fields agree at approximately  $1e-12$  relative L2 for stress, reactions and crystal slips, and at  $1e-16$  for displacement. Both used 192 substepped points. These are single-campaign measurements, not universal speed guarantees. The complete comparison is archived in `validation/_generated/performance/p43_m100_backend_comparison_latest.json`.

## 15. Current source notes and upstream design

The generated prototype uses  $S_{ref} = 210000$  MPa and the fixed 18-by-18 transport buffer described above. These should be replaced by a dimension-safe private transport and an explicit residual-scale contract before claiming a fully distributable generic backend.

The specialised source contains historical diagnostic comments about a three-per-thousand GPS/Schur tangent discrepancy. Those comments referred to a same-state transplant bug in an earlier diagnostic and are invalid as a qualification claim. The corrected result is that the one-step GPS tangent and the raw Schur agree when they start from the same physical state; the remaining Newton issue was the tangent of the host substepping composition.

Possible upstream designs are a Behaviour Brick, an extension of `StandardElasticity`, or a dedicated structural-plane-stress transformation in the `Implicit` DSL. The appropriate interface should expose the elastic block and its Jacobian before gradient subtraction, rather than relying on generated `fzeros` positions. It should also make the stress-dependence contract and residual scaling explicit.

## 16. Reproducibility and references

The reports record TFEL/MGIS and Python versions, source hashes, the EBSD source hash, orientations, parameter set, thread settings, Newton/Krylov diagnostics and field hashes. The independent condensation route, the specialised GPS route and the generic route are retained as separate validation artefacts.

Relevant MFront concepts are the `StandardElasticity`/Hooke stress-potential infrastructure, Behaviour Bricks, the generic behaviour interface and MGIS's batch material-data interface. See the official references:

- [Hooke stress potential and standard elasticity](#)
- [Behaviour Bricks](#)
- [Generic behaviour interface](#)
- [TFEL/MFront documentation index](#)

The general software reference is: T. Helfer, B. Michel, J.-M. Proix, M. Salvo, J. Sercombe and M. Casella, *Introducing the open-source MFront code generator: Application to mechanical behaviours and material knowledge management within the PLEIADES fuel element modelling platform*, *Computers & Mathematics with Applications* (2015), [doi:10.1016/j.camwa.2015.06.027](https://doi.org/10.1016/j.camwa.2015.06.027).

The repository's feasibility report records the TFEL 5.1 ABI probe and the generated-hook experiments: `validation/structural_plane_stress_mfront_feasibility.md`.

This note does not claim support for arbitrary MFront behaviours, finite strains, cohesive-zone behaviours, non-standard multi-gradient systems, or a distributed external Behaviour Brick for TFEL 5.1.